

Trajectory Improvement and Reward Learning from Comparative Language Feedback

Summary

December 23, 2025

1 Problem Setup

Traditional reinforcement learning from human feedback relies on pairwise comparisons between trajectories, which can be inefficient and limited in expressiveness. This paper introduces a novel approach that uses comparative natural language feedback to improve robot trajectories and learn reward functions. The goal is to enable more efficient and expressive human-AI collaboration through natural language communication.

2 Thesis

The central thesis is that comparative language feedback provides a more efficient and expressive alternative to pairwise trajectory comparisons for human-in-the-loop reinforcement learning. The authors argue that language allows humans to convey nuanced preferences and corrections that would be difficult to express through simple preference comparisons.

3 Key Concepts

Comparative Language Feedback: Natural language expressions that compare desired behavior to current behavior, such as "move faster" or "avoid the obstacle better."

Trajectory Encoder: A neural network that encodes robot trajectories into a latent space where they can be compared and modified based on language feedback.

Language Model: A pre-trained language model (T5) that processes human feedback and generates trajectory modifications in the latent space.

Co-finetuning: The process of jointly training the trajectory encoder and language model to align language and trajectory representations in a shared latent space.

Iterative Improvement: A process where humans provide feedback, the system modifies trajectories, and the cycle repeats until satisfaction is achieved.

4 Methods

The authors develop a system for learning from comparative language feedback:

1. **Trajectory Representation:** Robot trajectories are encoded into a latent space using a neural network that captures the essential features of the motion.

2. **Language Processing:** A pre-trained T5 language model processes human feedback and generates modifications to the trajectory representation.
3. **Latent Space Alignment:** The trajectory encoder and language model are co-finetuned to ensure that language feedback maps to appropriate trajectory modifications.
4. **Iterative Learning:** Humans provide feedback, the system generates improved trajectories, and the process continues until the human is satisfied.
5. **Reward Learning:** The system learns a reward function from the accumulated language feedback that can guide future trajectory generation.

The key innovation is the use of language as a direct interface for modifying trajectory representations, rather than just ranking alternatives.

5 Results

The authors demonstrate their approach on robotic manipulation tasks:

- **Efficiency:** Language feedback requires significantly less time than pairwise comparisons, with users achieving satisfactory trajectories in fewer iterations.
- **Expressiveness:** Users can convey nuanced preferences about speed, safety, efficiency, and task completion that would be difficult to express through simple comparisons.
- **Learning Performance:** The system successfully learns reward functions from language feedback that generalize to new situations.
- **User Preference:** In user studies, participants strongly prefer the language feedback approach over pairwise comparisons for both efficiency and expressiveness.
- **Co-finetuning Benefits:** Joint training of the trajectory encoder and language model significantly improves performance compared to using a frozen language model.
- **Real-World Applicability:** The system works effectively on both simulated and real robotic tasks, including kitchen manipulation and button pressing.

The results show that language feedback provides a more natural and efficient interface for human-AI collaboration in robotics.

6 Significance

This work introduces an important new paradigm for human-AI interaction in reinforcement learning:

- **Natural Interface:** Provides a more natural and expressive interface for humans to communicate preferences to AI systems.
- **Efficiency Gains:** Significantly reduces the time and cognitive effort required for humans to train and improve AI systems.
- **Scalability:** The approach scales to complex robotic tasks where traditional preference learning would be impractical.

- **Broader Applications:** The methodology extends beyond robotics to any domain where humans need to guide AI behavior through nuanced feedback.
- **Human-Centered AI:** Represents a step toward more human-centered AI systems that can understand and respond to natural language guidance.

The paper demonstrates that language can serve as a powerful bridge between human intuition and machine learning, enabling more effective collaboration and faster learning in complex domains.